

Ernesto Rojas

Curriculum Vitae

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EDUCATION

Stanford University

M.S. in Statistics

B.S. with Honors in Data Science

Subplan in Mathematical and Computational Science

Honors Concentration in Computational Neuroscience

Notation in Science Communication with Distinction (Portfolio)

Stanford, CA

2025 – 2026

2022 – 2026

RESEARCH EXPERIENCES

Research Assistant under Prof. Laura Gwilliams

June 2025 – Present

Gwilliams Laboratory of Speech Neuroscience

Stanford

Designed and wrote proposal for independent project to utilize the shared specialization between LLM embeddings and the brain during natural language processing to understand specific neural processes.

Applied and received internal grant funding for project, presented results at regional and national conferences.

Research Assistant under Prof. Russell Poldrack

June 2024 – September 2024

Department of Psychology and Statistics

Stanford

Evaluated statistical methods on removing motor artifacts from button pressing during fMRI tasks.

Replicated study on resting-state fMRI functional connectivity predictability to movie-watching fMRI data.

Research Assistant under Prof. Daniel Spielman

June 2023 – May 2024

Radiological Sciences Laboratory

Stanford

Implemented structure-guided total variation regularized image reconstruction with fast gradient projection for deuterium metabolic imaging scan enhancement.

Presented results at international radiology conference.

CONFERENCE PRESENTATIONS

Neuroscience 2026, Society for Neuroscience

Poster, In-Person

DC, 2026

Dynamics of Syntactic Resolution in the Inferior Frontal Gyrus During Natural Speech Comprehension

AI+Science: Accelerating Discovery Conference, Stanford HAI and Data Science

Poster, In-Person

Stanford, 2026

Attention-Guided Encoding Models Uncover Syntactic Computations During Natural Speech Comprehension

Annual Meeting, International Society for Magnetic Resonance in Medicine

Digital Poster, In-Person

Singapore, 2024

Evaluation of Anatomical Guided Reconstruction for Improving the Spatial Resolution of Deuterium Metabolic Imaging

GRANTS & AWARDS

Awards

NSF GRFP Honorable Mention

National Science Foundation

Departmental Honors

Department of Data Science, Stanford University

Award of Excellence

Stanford Alumni Association

Honors graduating seniors who have demonstrated a sincere commitment to the university through involvement, leadership, and extraordinary Stanford spirit.

Grants

Trainee Professional Development Award — \$1,000 (*Pending*) 2026

Society for Neuroscience

Major Grant — \$9,500 2025

Stanford Vice Provost for Undergraduate Education

Major Grants support student-proposed, full-time research projects supported by a faculty mentor.

Conference Travel Grant — \$1,500 2024

Stanford Vice Provost for Undergraduate Education

ISMRM Educational Stipend — Waived Fees (\$237) + \$300 2024

International Society for Magnetic Resonance in Medicine

Summer Undergraduate Research Program Stipend — \$9,000 2024

Department of Statistics, Stanford University

Research Experience for Undergraduates Program Stipend — \$9,000 2023

Radiological Sciences Laboratory, Stanford University

Total Amount Awarded: \$29,300

TEACHING AND ADVISING

Teaching Assistant Winter 2024, Summer 2024, Winter 2025, Winter 2026

Principles of Data Science Stanford, CA

Led section for 18–24 students twice a week teaching fundamental data science skills. Organized and held weekly 3-hour long office hours for "Data Science Nights". Created review materials and held review sessions before exams. Redesigned assignments to integrate data storytelling content.

Data Science Peer Advisor 2025 – 2026

Department of Data Science, Stanford University Stanford, CA

Held weekly office hours to advise prospective data science students on course selection and other questions related to the major. Organized community events and held talks on applying data science to research.

RELEVANT PROJECTS

Deep Learning Decoding Models for Active Speaker Recognition during Movie-watching March 2025

Trained and evaluated LSTM, RNN, and GRU architectures on a large-scale movie-watching neural dataset to classify current speaker, achieving 80% accuracy and exceeding the state-of-the-art results.

Analyzed predictions to guide targeted architecture optimizations, improving model robustness.

Boosting Neural Decoders with RL Fine-Tuned LLMs during Natural Speech Comprehension May 2025

Integrated an RNN-based neural decoder with a fine-tuned LLM, boosting decoding accuracy 23% above baseline on held-out neural test sets.

Created synthetic chain-of-thought datasets and fine-tuned Llama 3.2 using SFT, DPO, and RLOO, enhancing reasoning quality and task-specific performance.

Hierarchical Bayesian Models for Real-Time Election Calling

May 2026

Designed, built, and evaluated hierarchical Bayesian models to quantify uncertainty around California congressional races during the 2024 midterm election.

Investigated and corrected sources of systemic bias and validated confidence bands for accurate quantification.